



Vijay Academy Presents



Savitribai Phule Pune University

Web Technology (WT)

TE Computer Engineering

Endsem Exam Strategy (Unit 3-6)

PYQs + Most IMP Questions

What You Get : Strategy For Exam :

- Simple Explanation
- PPT + Notes
- PYQs + Most IMP Questions



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UNIT 3 — Java Servlets and XML

MOST IMPORTANT TOPICS (Must Study)

1. Servlet Life Cycle ---Asked in almost every paper — never skip this!

Explain servlet lifecycle.

Explain servlet lifecycle with session management using cookies and URL rewriting.

What is Servlet? Explain lifecycle with example.

2. Session Management — Cookies & URL Rewriting ---Asked in every paper in different forms.

How cookies & URL rewriting for session management.

Explain session management using cookies and URL rewriting.

How cookies are used to track a session.

3. DTD in XML — Schema, Elements, Attributes ---Asked in every paper — very high frequency!

What are DTDs? Explain how they work

Explain DTD with schemes elements attributes

Differentiate XML DTDs vs XML Schema

Distinguish between external and internal DTDs



4. doGet() vs doPost() Methods ---Asked in almost every paper.

Compare doGet and doPost methods

Explain doGet and doPost methods. Differentiate doGet vs doPost

Write servlet which accepts two numbers using POST method

5. AJAX ---Asked in every single paper without fail!

Write short note on AJAX

What do you understand by AJAX? Explain it

Write note on AJAX

6. Servlet Architecture ---Asked in almost every paper.

Describe servlet architecture in detail

Explain servlet architecture with diagram

Explain servlet architecture in detail



MEDIUM IMPORTANT TOPICS:

1. XML — Documents, Declaration, Namespaces ---Asked in multiple papers.

What are strengths of XML technology? Explain need of XML

Explain XML with respect to structure, declaration syntax, namespace

Write note on XML

2. XML Transformation ---Asked in a few papers.

Write short note on XML transformation

Explain process of transforming XML document

3. Servlet with MySQL Database ---Asked in multiple papers.

Explain servlet and MySQL database connectivity with example code to display data from employee table

4. HTTP Session ---Asked in a few papers.

What is session? List session tracking techniques

HTTP session — explain

5. DOM Based XML Processing ---Asked in a couple of papers.

Explain DOM based XML processing



LEAST IMPORTANT TOPICS

1. Servlet Generating Dynamic Content
2. Parameter Data in Servlets
3. Servlets Concurrency
4. Data Storage in Servlets
5. Hello World Servlet

Unit 4. JSP and Web Services

MOST IMPORTANT TOPICS (Must Study)

1. JSP Life Cycle ---Asked in almost every paper — very high frequency!

Explain life cycle of JSP

Explain JSP lifecycle with diagram

Explain JSP lifecycle. Differentiate JSP vs Servlet

2. Struts Framework — Architecture, Actions, Interceptors ---Asked in every paper without fail!

Explain struts framework with its components. Also explain interceptors

Explain struts framework with respect to architecture, actions, interceptors & exception handling

Write note on struts. Explain struts with example code block

3. WSDL and SOAP ---Asked in every single paper!

Write short note on SOAP and WSDL

What is WSDL and SOAP? Explain WSDL in detail

Explain concept of WSDL and SOAP

Write benefits of web services and explain SOAP, REST and UDDI



4. JSP Support for MVC ---Asked in multiple papers.

Explain JSP support for MVC paradigm

Draw and explain MVC architecture for developing web application

Explain JSP with support for Model View Controller

5. Web Services — Concepts and Benefits ---Asked in every paper.

Identify and justify benefits of using Web Services

Explain concept of web services

Write benefits of web services

6. JSP vs Servlets ---Asked in multiple papers.

List and write advantages of JSP over servlet

Explain JSP vs Servlets (Min 4)

Explain how JSPs are translated to Servlets



MEDIUM IMPORTANCE TOPICS

1. JavaBeans Classes and JSP ---Asked in a couple of papers.

Explain JavaBeans classes and JSP

Explain JavaBeans classes and JSP with example

2. JSP Basic Syntax — Directives, Scriptlets, Expressions ---Asked in multiple papers.

Write a JSP program to demonstrate use of page directive, Scriptlet, Expression and Comment

Explain concept of JSP with syntax and sample example

3. JSP Implicit Objects ---Asked in one paper.

List and elaborate any five JSP implicit objects with example

4. Writing a Java Web Service ---Asked in a couple of papers.

Explain concept of JSP with syntax and sample example. Explain analogy of JSP and Servlets

5. JSP Related Technologies ---Asked rarely as standalone.

Appeared as part of JSP introduction questions only



LEAST IMPORTANT TOPICS

- 1. Running JSP Applications**
- 2. Web Service Client**
- 3. Struts — Result Types, Localization**
- 4. Struts — Annotations**
- 5. Struts — Validations**

Unit 5. Server Side Scripting Languages

MOST IMPORTANT TOPICS (Must Study)

1. Arrays in PHP ---Asked in every single paper — highest frequency topic!

What is multidimensional array in PHP? Explain with simple PHP code

Explain various types of arrays in PHP with example code

Explain PHP with respect to Arrays

2. MySQL with PHP ---Asked in every paper without fail!

Identify and explain steps involved in connecting MySQL with PHP

Explain object oriented way to connect MySQL database with PHP

Write PHP code to connect MySQL database to display records

Explain MySQL and PHP for inserting student data

3. Session Tracking in PHP ---Asked in multiple papers very frequently!

Explain how cookies and session are used for session management in PHP

Session tracking in PHP

Explain functions in PHP with example and session management

4. Cookies in PHP ---Asked in multiple papers.

Explain how cookies and session are used for session management in PHP / Explain cookies in PHP with example



5. WAP and WML ---Asked in almost every paper!

Explain in detail WAP Architecture and WML

What is WAP? Explain components of WAP architecture in detail

What is WML? Explain WML elements

Explain WAP and WML

Write short note on WAP and WML

6. Overview of .NET Framework ---Asked in every paper!

Overview of ASP.NET

Draw and explain .NET framework with CLR, CLI

Explain concepts of WAP, WML and .NET framework

7. Functions in PHP ---Asked in multiple papers.

Explain functions in PHP with example

Explain PHP with respect to Arrays, Functions, Control statements

How does this function work in PHP? Explain with example

Explain PHP functions and arrays in PHP



MEDIUM IMPORTANCE TOPICS

1. Control Statements in PHP ---Asked in a few papers.

Explain PHP with respect to Arrays, Functions, Control statements

2. Overview of C# ---Asked in a couple of papers.

C# vs Java (Min 4 points)

Explain concept of .NET framework and C#

Write note on ASP.NET and NodeJS

3. Overview of Node JS ---Asked in multiple papers.

Explain overview of Node JS

Write note on ASP.NET and NodeJS

4. Introduction to ASP.NET ---Asked in a few papers.

Explain overview of ASP.NET

5. Form Handling in PHP ---Asked indirectly in multiple papers.

Explain MySQL and PHP for inserting student data(form used)

Explain steps involved in connecting MySQL with PHP(form handling used)



LEAST IMPORTANT TOPICS

- 1. Introduction to PHP and Uses**
- 2. General Syntactic Characteristics**
- 3. Primitives, Operations and Expressions**
- 4. Output in PHP**
- 5. Files in PHP**
- 6. ASP.NET Controls**
- 7. Web Services in ASP.NET**

Unit 6. Ruby and Rails

MOST IMPORTANT TOPICS (Must Study)

1. Scalar Types and Their Operations in Ruby ---Asked in almost every paper — highest frequency topic in Unit 6!

Explain scalar types and their operations in Ruby

Explain Ruby with its advantages. Explain scalar types

Explain Ruby with respect to scalar types, control statements and arrays

2. Arrays in Ruby ---Asked in every paper without fail!

Explain types of arrays in Ruby. Explain array operations in Ruby

Explain concept of classes and arrays in Ruby

Explain Ruby with respect to scalar types, control statements and arrays

3. Rails with Ajax ---Asked in almost every paper!

Explain Rails with AJAX in detail

Explain Rails applications and databases. Also explain Rails with AJAX

Explain concepts of Rails with AJAX and EJB



4. EJB — Architecture and Types ---Asked in every single paper without fail!

Explain architecture of EJB and explain types of EJB

Draw and explain role of EJB container in enterprise applications

What is EJB? Draw and explain main components of EJB architecture

5. Control Statements in Ruby ---Asked in multiple papers.

Explain how multiple selection constructs are implemented in Ruby

Explain Ruby with its advantages. Explain control statements in Ruby

Explain Ruby with respect to scalar types, control statements and arrays

6. Classes in Ruby ---Asked in multiple papers.

Explain classes and objects in Ruby with appropriate examples

What is a module? Explain difference between classes and modules

Explain concept of classes and arrays in Ruby

7. Overview of Rails ---Asked in multiple papers.

Introduce concept of Rails. Describe layouts and stylesheets in Rails

What are positive aspects of Rails, explain with example

Explain Rails applications and databases



MEDIUM IMPORTANCE TOPICS

1. Rails Applications and Databases ---Asked in a couple of papers.

Explain Rails applications and databases. Also explain Rails with AJAX

Explain Rails applications and databases

2. Document Requests in Rails ---Asked in a few papers.

Explain document requests in Rails

Explain document requests and processing forms in Rails

3. Processing Forms in Rails ---Asked in a couple of papers.

Explain document requests and processing forms in Rails

4. Layouts in Rails ---Asked in one paper.

Introduce concept of Rails. Describe layouts and stylesheets in Rails

5. Methods in Ruby ---Asked in a couple of papers.

Explain how to write the methods and call the method in Ruby with example

6. Pattern Matching in Ruby ---Asked in one paper.

Explain scalar types, operations and pattern matching in Ruby



LEAST IMPORTANT TOPICS

- 1. Hashes in Ruby**
- 2. Introduction to Ruby — Origins and Uses**
What is Ruby? List some features of Ruby" — asked once only
- 3. Simple Input and Output in Ruby**
- 4. Code Blocks and Iterators in Ruby**



Thank You!



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